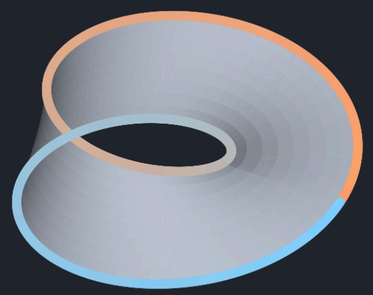
Pi & Planck

TOPOLOGY

The Strip With One Side

Give a paper strip a half-twist and tape the ends: it now has one side and one edge. Möbius and Listing found it in 1858.

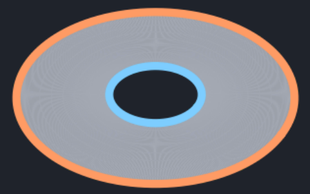
THE MÖBIUS STRIP



One edge: follow it once round and you are on the far side, so it takes two laps to close.

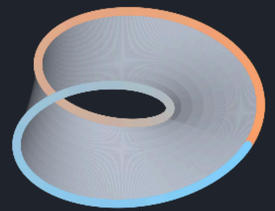
1. Take a strip of paper
2. Add one half-twist (180°)
3. Tape the ends together

COUNT THE HALF-TWISTS



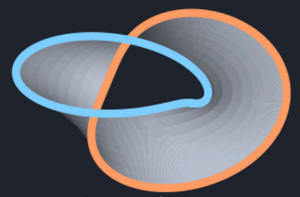
0 twists

2 sides · 2 edges



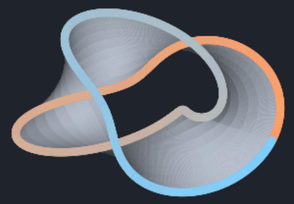
1 twist

1 side · 1 edge



2 twists

2 sides · 2 edges



3 twists

1 side · 1 edge

CUT DOWN THE MIDDLE

0 twists	→ Two thin bands
1 twist	→ ONE band, twice as long
2 twists	→ Two linked bands
3 twists	→ ONE band, tied in a knot

1858: found twice

August Möbius and Johann Listing discovered the one-sided strip independently, in the same year.

Born 17 November 1790

August Ferdinand Möbius was born on this day in Schulpforta, Saxony, and worked as an astronomer and mathematician in Leipzig.

1 side: you can paint the whole strip without lifting the brush

1 edge: one loop, not two

2× as long: the strip cut down the middle becomes one band

Sources: MacTutor History of Mathematics, biographies of A. F. Möbius and J. B. Listing (University of St Andrews) · A. F. Möbius, Berichte Sächs. Ges. Wiss. 17 (1865) 31–68 · Edge and cut counts checked by computation for this graphic